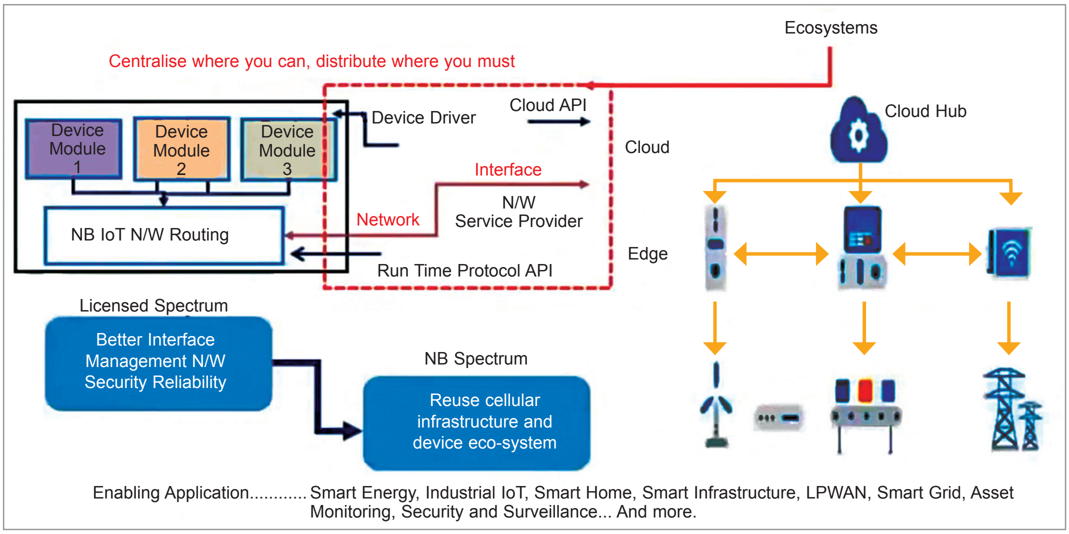

Fig. 8: NBIoT deployment

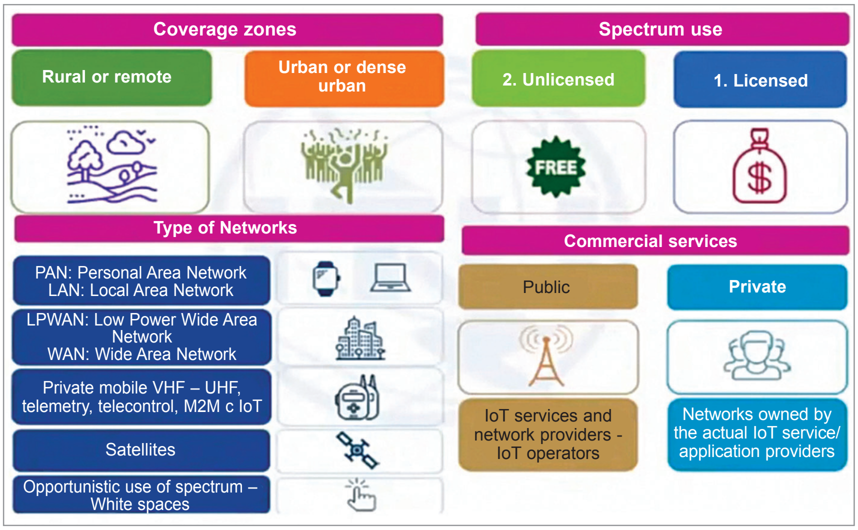

Fig. 9: Use cases

Long battery life. Up to ten years of operational lifetime with a 5-watt-hour battery (depending on traffic and coverage needs).

Supports high number of devices. At least 150k per cell.

Three modes of operation. These are:

1. Stand-alone. Spectrum currently used by GERA N (GSM Edge Radio Access Network) systems as a replacement of one or more GSM carriers.

2. Guard band. Unused resource blocks within an LTE carrier's guard band.

3. In-band. Resource blocks within a normal LTE carrier

In LTE carrier, we can use 3G or 4G. We have to use different modulations on the same band. And for that modulation technique we require standalone operation, as well as we are sampling a frequency modulation that is either guardband or sideband. The difference between them is that the edge of a signal (for any spectrum) is always the sideband. And the amplitude of the middle of the signal is the guardband. The modulation is 200kHz.

The architecture of NBIoT is similar to the GSM architecture. But there is a difference in PDN gateways, serving gateways, and even the end devices. Take for instance LTE. We are targeting end devices that will go for NBA as well as for PDN gateways. But here on the LTE network, all the transmitters and receivers are on the side of the network. We need to make a change on the gateway (that is similar to the PDN gateway and server gateway). And the network service provider is providing maximum signal availability to the infrastructure.

The challenge of Narrowband is that the network is not available in the market. Mobile network operators (MNOs) are working on this. So, on the design side, we are working with Narrowband Fallback 2G since 2G is available everywhere in the market. Wherever Narrowband is not available, the module will work on 2G. This altogether depends on standards, infrastructure, and mobile data rules.

Challenges and solutions

Some challenges being faced, and the solutions suggested for them are listed below.

Effect of resources. Maintenance of networks from threats. MNOs are going to change the network on the transmitter and receiver side for 2G, 3G, and LTE network as per the Narrowband.

Security. Hardware-level security needs to be defined when changing from 2G, 3G, or 4G. Cryptography is solving security and privacy issues.

Control access. Policies need to be defined for device control and user access.

Interoperability. Integration flexibility with device driver IP protocol.

Legality and rights. Data protection laws to be followed.

Economy and deployment. Investment incentives and technical skill required.

Considering the growing number of IoT devices and the requirement for low power, Narrowband IoT is the way forward as it meets this demand and offers fast connectivity along with robust and secure deployment. EFY

The article is based on the talk 'Telecom Narrowband IoT: The Next Big Thing' by Dharmendra Kumar, Head of IoT at Arihant Group, presented at February edition of Tech World Congress 2021. Vinay Prabhakar Minj is a technical journalist at EFY.